

User's Guide and Reference Manual for

SimUaidTM

Software for Logic Simulation

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Preface

SimUaid is a computer program you can use to simulate logic circuits composed of gates, flip-flops, adders, registers, counters, and other basic logic components. Using SimUaid, you can input a logic diagram and display it on the computer screen. After inputting the diagram for a logic circuit, you can simulate its operation by applying 0's and 1's to the input pins and observing the circuit outputs on the screen.

Live simulation mode allows immediate observation of responses to input switch changes. Synchronous simulation mode allows stepping one clock period at a time to observe changes in device inputs and outputs on the screen. In this way you can verify that your design is correct before actually building it in lab.

The *User's Guide and Reference Manual for SimUaid* describes how to use the SimUaid logic simulator. It describes the basic SimUaid operations, including entering and opening a circuit file, editing the circuit, simulating the circuit, viewing the waveform, printing the circuit and waveform, and saving the circuit to disk.

SimUaid was originally developed for use in logic design classes at the University of Texas. The software was developed for use with *Fundamentals of Logic Design*, 5th ed., by Charles H. Roth (Brooks/Cole, 2003). However, it can be used with almost any logic design textbook.

SimUaid User's Guide

Note on Component Size: Depending on the operating system and configuration of your computer, you may experience some problems displaying the parts on the screen. If the parts do not appear properly on your screen, change the graphics mode. You can do so by selecting the Change Graphics Mode command on the View menu.

Refer to section 1.3.8, *Change Graphics Mode*, in the *SimUaid Reference Manual* for more information about how these modes work.

1. Entering a Logic Circuit

Start with a new circuit window by selecting New from the File menu or by clicking the  button.

1.1. Placing a part

To place parts onto the circuit window, click on the Parts menu and select the desired part. The mouse pointer will be changed into the symbol of the selected part. Move the mouse over the circuit window and place the part in the desired location by clicking the left mouse button. The same part will continue to be placed until you select another part or action.

1.2. Connecting the wires

To connect a wire from one part to another, first select the Wire option from the Edit menu, or simply click the  button from the toolbar at the top of the circuit window. The mouse pointer becomes a cross. Move the center of the cross to an input or output pin of a part and click on it. A beep will sound if the wire is connected to that pin. If there is no beep, this means that the wire is not connected to anything. After clicking on the pin, you can click on anywhere on the circuit window to establish a corner point for the wire. To establish a connection to the next part, just click on the input or output pin of the part. Again, you will hear a beep if the wire is connected to the pin of the part.

When drawing the connecting wires, be careful to avoid unnecessary crossed wires. This will help minimize the number of errors, and it will be easier to debug the circuit. The following are examples of good and bad connections. Note that when two wires touch each other, there is no connection unless a black dot appears at the intersection.

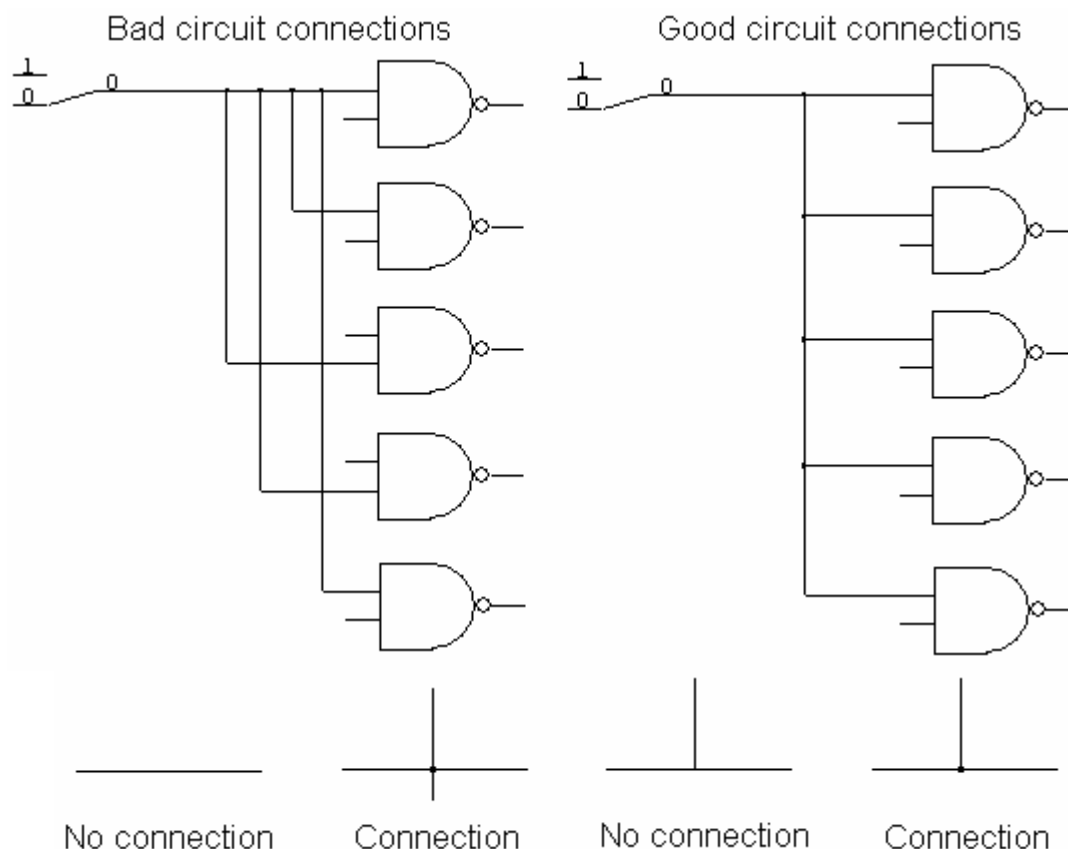


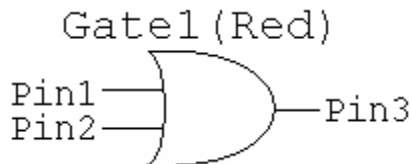
Figure 1. Circuit Connections

You can verify all of your circuit connections by the following procedure:
 Move the mouse pointer over a wire near the output or input of a part and click while holding the Shift key down. All the connections to that wire will be highlighted as a red dashed line. You can repeat this procedure for each group of wires to ensure that all the wires are connected to the right places.

1.3. Labeling parts and pins

You can label each part by clicking on the Label option on the edit menu or by clicking the  button. The mouse pointer will change into a pencil. Move the pencil point over the middle of the part you want to label and click. Enter the desired label, and then hit Enter. The part label is red and will be placed above the part. To label a pin, follow the same procedure, except click on a pin. The pin label is black and will be placed beside the pin.

Example: In the figure below, Gate1 is a part label; Pin1 and Pin3 are pin labels.



Note: To show the values of input and output pins, select Show I/O Values from the View menu, or click on the Show I/O Values button .

1.4. Selecting devices and labels

Clicking on a wire segment highlights only that segment; however, clicking on a wire segment while holding down the Shift key highlights the entire net which contains that segment. This helps you to observe the interconnection of wires, and know which is connected to which.

When a device is selected, its label is also selected. However, when a device's label or a pin's label is selected, only the label is selected, not the entire device.

To select the entire circuit, choose Select All from the Edit menu.

1.5. Deleting devices and wire segments

Choose Select on the Edit menu or the  button on the toolbar. Move the tip of the Pointer to the center of a device or wire segment and click. Then hit the delete key or click the  button.

1.6. Moving a device

Select Move on the Edit menu or the  button on the toolbar. Position the hand in the middle of the device or gate to be moved. Hold down the mouse button and drag the device or gate to the desired position. The connecting wires will follow along. Note: If the program will not allow you to move a device, it is probably because moving the device would make the diagram difficult to read. To move a device a small distance, you can use the "nudging" feature. First click the  button on the tool bar, then select the device you want to move, and use the four arrow keys on the keyboard to move it.

1.7. Moving a wire or label

The procedure is similar to that for moving a device. To move an entire circuit, click on the  button, and then click on Select All on the Edit menu.

1.8. Deleting a group of wires (called a net)

Choose Select on the Edit menu or the  button on the toolbar. Move the tip of the pointer to a wire in the net to be deleted. Click on the wire while holding down the Shift key. This will highlight the entire wire net to be deleted. Then hit the delete key or select delete from the Edit menu or the scissors icon on the toolbar.

1.9. Input and output values

The input and output signals can have four different values:

0	Logic 0
1	Logic 1
Z	The terminal does not have an input, i.e. no connection was established.
X	The output is undetermined because an input to the part was undetermined or not connected.

To show the values of input and output pins, select Show I/O Values from the View menu, or click on the Show I/O Values button  or press Ctrl-i.

Figure 2 illustrates the different values of input and output for a logic circuit. Note that the connection between the switch and the wire to the top input of the AND gate is bad because there is no black dot at the intersection. Therefore, the input to the AND gate is Z, and the output of the AND gate is X.

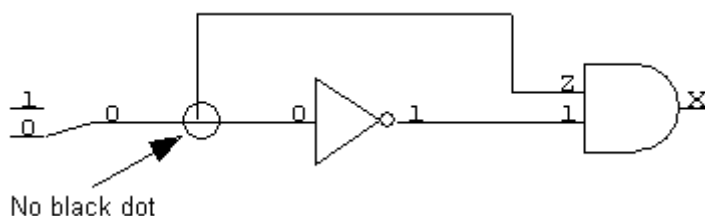


Figure 2. Circuit with Bad Connection

1.10. Example for entering logic circuit

The following procedures will give you an idea of how to enter a simple logic circuit. As an example, enter the following diagram into the circuit window:

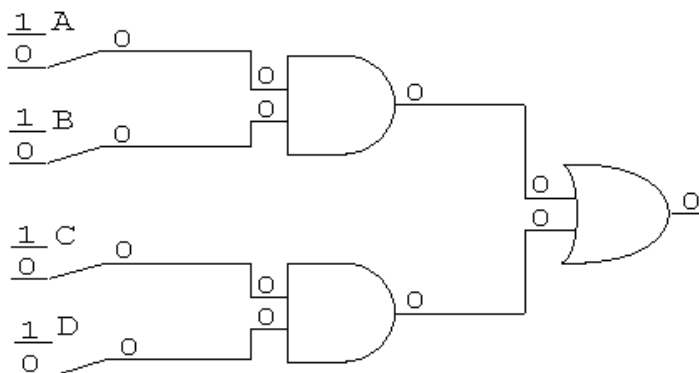


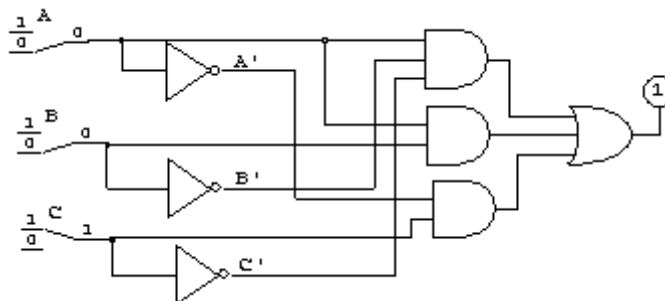
Figure 3. Example Circuit Diagram

- a. First, place the gates on the circuit diagram. To do this, click on the parts menu and select the *AND-2 input* (2-input AND) gate from the submenu. Use the mouse to position the first *AND* gate where you want it and click to place it. Use the mouse to position the second *AND* gate and click to place it.
- b. Then click on the parts menu and select the *OR-2 input* (2-input OR) gate on the OR submenu. Use the mouse to position the *OR* gate and click to place it.
- c. Next click on the Parts menu and select *Switch>>right*. Use the mouse to place four switches in the desired locations on the screen.
- d. To display the logic levels of each pin, select *Show I/O Values* from the View menu or click the  button. The logic levels {1, 0, Z, or X} will appear above the input and output pins of each component in the circuit.
- e. Next, connect the pins of the parts with wires. The general procedure for drawing a wire on the screen is as follows:
 - i. Click on the *Wire* option from Edit Menu (or the  button). The pointer becomes a cross hair.
 - ii. Click the left mouse button at the starting point of the wire.
 - iii. Move the mouse to the ending point, and click there. If you wish to change the routing of the wire you may click at additional intermediate points as you draw the wire.
- f. If you need to move a gate or device, select the *Move* option from Edit menu or the  button. Position the mouse to the middle of a gate or device and drag it to its new location. The connecting wires will follow along.
- g. Next we will label the switches. Select *Label* in the Edit menu or click on the  button. Note that the cursor becomes a pencil. Move the tip of the pencil to the middle of the top switch and click. Type in the label *A* followed by Enter. Now use the pencil tip to select the next switch, and type in the label *B*. In a similar manner, label the next two switches *C* and *D*.
- h. Your circuit is now complete and should look similar to the one in Figure 3. You are ready to simulate your circuit.

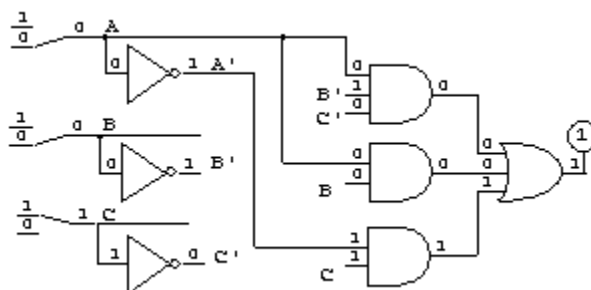
1.11. Using Virtual Connections

The example given below explains the difference between wired and virtual connections. If you enter all of the connecting wires in a SimUaid circuit diagram, the diagram may become very cluttered and hard to read. The virtual connection feature in SimUaid allows you to connect two device pins together without actually drawing the wire. To make virtual connections, place a label on a device output pin. Then place the same label on one or more device input pins. The output will be connected to the inputs, even though the connecting wires are not shown on the screen.

The following example shows a logic circuit with no virtual connections:



The same circuit is redrawn below, with wired connections for A and A', and virtual connections for B, B', C, and C'. In order to make this work, B, B', C, and C' must be pin labels. Using B' as a pin label does not actually complement B; you still must use an inverter with a B' output. To label a pin, first click on the ABC button on the SimUaid menu. Then use the pencil cursor and click on a device pin (not on the device itself). Type the label and hit return. The label should appear in black near the pin. Then you may move the label anywhere you want it, but do not move it too far away from the pin.



After you have input your circuit, select Show I/O values on the View menu. Every device input should be 0 or 1. If any input or output is X or Z, you have a bad connection, and you should fix this before continuing. In the above diagram, note that the switch output pin labeled B has a 0 value, and the gate input pin labeled B has a 0 value, even though no visible wire appears between the two pins. Similarly, the pins labeled B' both have a 1 value.

1.12. Redraw Window

You might want to redraw the window occasionally if your circuit looks like it is not displayed correctly, or if your changes do not seem to take effect. To redraw the window, select Redraw Window from the View menu.

1.13. Scaling and printing your circuit

Before printing your circuit by selecting Print from the File menu, ensure that the output will look the way you want it to. You might want to fit the design on a certain number of pages, or if it must use two pages, you might want it to flow to the right instead of down. The vertical and horizontal dotted, red lines you see on the circuit window are page boundaries. To keep your circuit on one page, you must fit it within the left, uppermost rectangle.

To check if your output is flowing to the next page, you can check the Print Preview from the File menu. The print preview shows how your design will look when printed on one or more pages. Click on the Close button to Close the Print Preview and return to the design.

If your output is too large or too small, you can select Set Scaling Factor from the View menu to make your circuit the desired size. Select Fit to Window from the View menu to scale the circuit to just fit in the visible window and on one printed page.

1.14. Adding comments

To add a comment to your circuit, select **Comment** on the **Edit** menu, type in your comment, and then press the F12 key.

2. Simulating the Logic Circuit

You can simulate your circuit manually or by using the clock and input signals.

2.1. Manual Simulation

To manually simulate your logic circuit, you will need to connect switches to the inputs of your logic circuit. Do so by selecting Switch from the Parts menu and placing them on your design window. To observe the outputs of the logic circuit, you can place probes on the output pins of your logic circuit. You can also select the Show I/O Values from the View menu to make SimUaid display the input and output values of all the parts in the design window.

To toggle the switch between logic 0 and 1, click on the  icon or press Ctrl+S to switch to the select mode. Then, click on the middle of the switch. This will cause the switch to toggle. To simulate the circuit, just toggle the switches and observe the output values from the probes. The output values will change to reflect the different input signals that are being fed into your circuit.. (If the output values do not change immediately, reset the simulator by clicking on the Reset option on the Simulate menu or the  button. If this does not fix the problem, you may have accidentally quit the Live Mode. In the "Simulate" pull-down menu, be sure that "Change Mode" does NOT have a check mark next to it. The Change Mode allows advanced users to analyze circuit timing. It is discussed in Appendix B of the *SimUaid Reference Manual*.)

Example:

Following the example in the previous section, click on the  button to change to the Select mode. Then set the switches to $A = 0$, $B = 1$, $C = 1$, and $D = 1$. You can do so by clicking on each switch until the desired input is obtained. Then, observe that the circuit output is logic 1, as shown in the Figure 4.

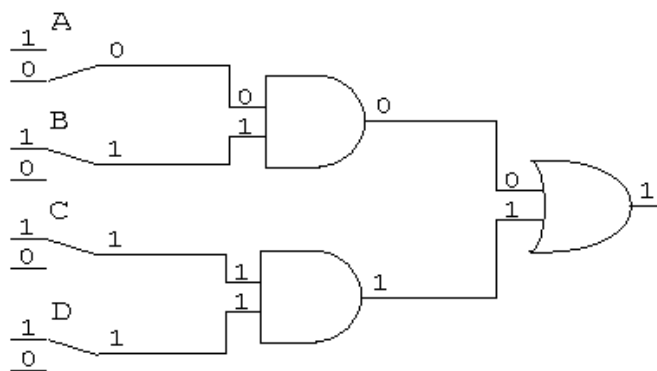


Figure 4. Example Circuit Diagram

Continuing in this manner, you can observe the outputs of the circuit for the desired input combinations and see if it works correctly.

2.2. Simulating Flip-Flops with SimUaid

SimUaid provides two types of flip-flops on the Parts menu—a J-K flip-flop which changes state on the falling edge of the clock input, and a D flip-flop which changes state on the rising edge of the clock input. Work through this exercise to become familiar with using a D flip-flop.

- Start with a new empty circuit window on the screen. Select the D flip-flop from the Parts menu and position it in the center. The clock input at the lower left is marked with a small triangle. This flip-flop changes state at the rising edge of the clock signal. Also note that the active-low Preset and Clear inputs are labeled "S" and "R" respectively.
- SimUaid requires that a signal be connected to every flip-flop input **including the preset and clear (S and R) inputs**. Connect switches to the flip-flop inputs and outputs as shown in Figure 5.

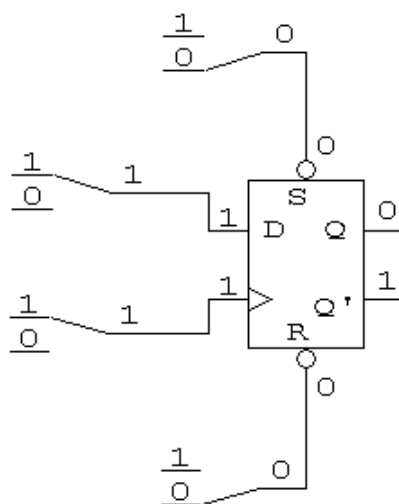


Figure 5. D Flip-Flop

- c. Click on the switches as required to set all inputs to 0 and then set the D input to 1. Next, input a rising edge on the $CLOCK$ by changing the switch from 0 to 1. Why did the flip-flop state Q remain at 0? Because having both Preset (S) and Clear (R) equal to 0 at the same time is an illegal input combination. Also, note that the Preset and Clear inputs are *asynchronous* inputs; they determine the flip-flop state regardless of the clock. Correct this condition by setting Preset (S) to 1 and then Clear (R) to 1. At this point you should have $Q = 0$ since the Clear input was 0 (and so active) briefly before it was set to 1. If you input another rising edge at the $CLOCK$ input, the flip-flop will now change to $Q = 1$ in response to the D input.
- d. Verify the operation of Clear (R) by setting it to 0 and then back to 1. Then verify Preset (S) in the same way.
- e. Starting with $Q = 0$, $D = 0$, and $CLOCK = 1$, change the clock input to 0 and back to 1. Note that Q will remain 0. Now change D to 1 and change the clock to 0 and back to 1. Q will change to 1. (Note that the flip-flop changes state just after you change the clock from 0 to 1). Then set $D = 0$ and toggle the clock again. When does the flip-flop change state?

2.3. Automatic Simulation

Automatic simulation is usually performed for sequential circuits, for example circuits involving flip-flops. To automatically simulate your logic circuit, you will need to insert a Clock and an Input Signal in your circuit design. The Input Signal will go in place of the switches of the input to your logic circuit. Figure 6 gives an example of a sequential logic circuit using automatic simulation. This circuit is named `Mealy_ex.ckt` and is found in the `SimUaid/examples` folder.

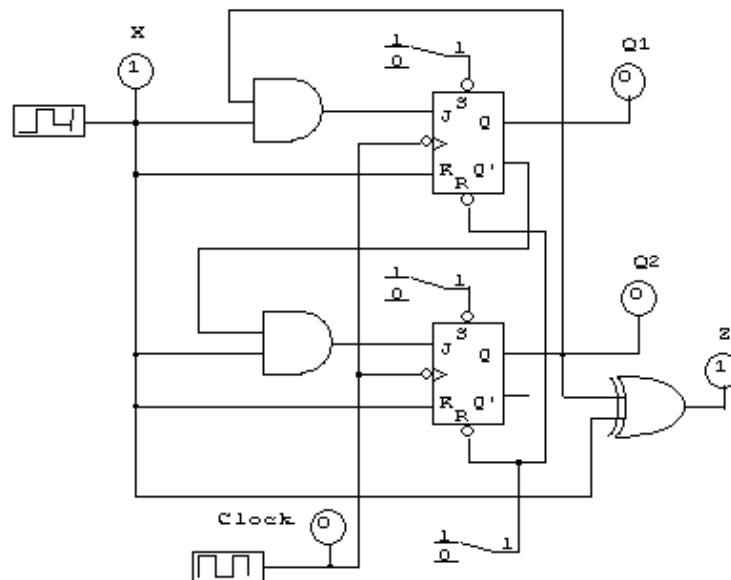


Figure 6. Example for Automatic Simulation

2.3.1. Configuring timing aspects for automatic simulation

Configure the Clock and Input Signal by clicking on the  icon on the tool bar and then clicking on the Clock or the Input Signal. Clicking on the clock will bring up the Simulation Options dialog box. Here, you can configure the period of the clock, the initial value of the Clock output, and the time limit of the simulation. Figure 7 below shows the simulation options dialog box.

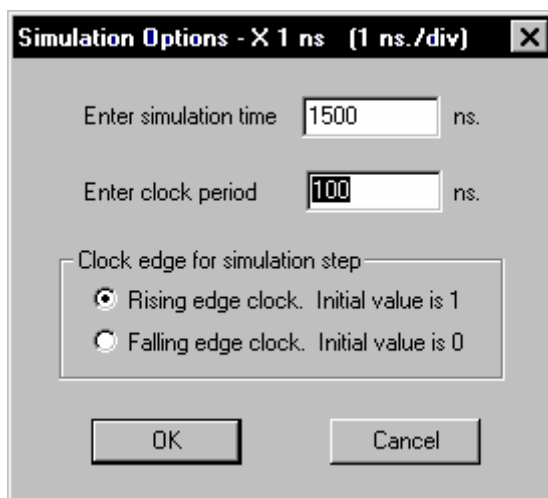


Figure 7. Simulation Options

The default propagation delay for all parts is 1 ns. You might want to change the delays for individual parts, if it is required for your simulation. To do so, click on the  icon or select Delay from the Edit menu, then click on the part you want to set a special delay for. The Propagation Delay dialog box appears so you can enter the delay for the part in ns.

You can change the default delay for all parts by selecting Default Delay from the Edit menu. After you have entered the default delay, all the devices you place on the circuit will have this propagation delay until you change the default delay again. Devices that are on the circuit before this command is executed will not be affected.

The input signal can be set up as shown in Figure 8. The values are for a sequence 1100 1011 01. To enter new values, enter the Time in ns and the associated input signal value. Then click Add/Modify. To delete a value, click on the row of the value and click Delete. The values can be 1, 0, X, Z, or R. For more information, refer to the *SimUaid Reference Manual*.

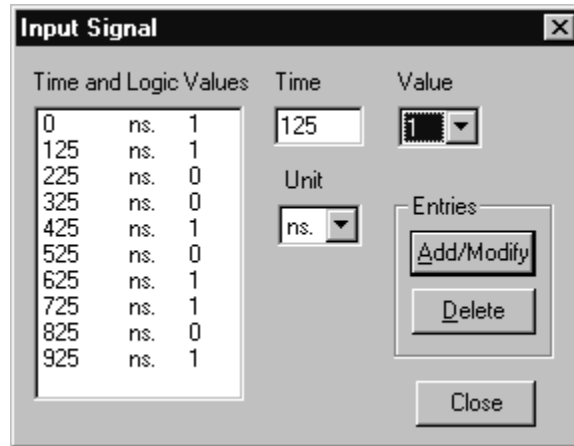


Figure 8. Input Signal Configuration

The outputs of the simulation are displayed in a timing waveform chart. Click on the Open Timing Diagram command on the Window menu to bring up the timing chart.

Start the simulation by clicking on the Go option in the Simulate menu. This will make the simulation run until it reaches the time limit. The Step command can be used to run the simulation for one clock period. The Change command will run the simulation until a value changes. The simulation can be stopped at any time by selecting the Stop command.

The outputs of the simulation are displayed in a timing waveform chart. Click on the Open Timing Diagram command on the Window menu to bring up the timing chart.

You will need to place probes on the signals that you want to display on the timing diagram. The order in which you place the probes will be the order of the timing waveforms. Label each probe with the appropriate name so that the label will appear next to the probe's waveform in the timing window. Probes without labels will have "???" appearing next to their timing waveform.

Figure 9 shows an example of a timing diagram for the circuit shown in Figure 6. The horizontal scale of the waveforms can be adjusted by using the Scale option from the Options menu.

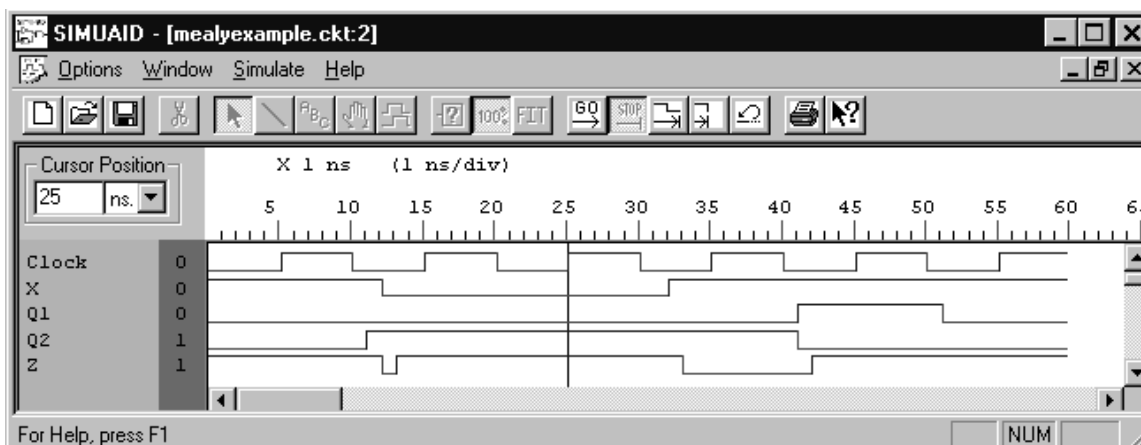


Figure 9. Timing Diagram Window

2.4. Debugging your circuit

SimUaid checks certain things about your circuit, such as whether pin or device names conflict and whether you try to connect a output to another output, but you must determine whether your circuit works according to your specifications, and correct any errors you may notice.

Here are some hints on how to debug your circuits.

- Make sure you know what the output should be in every case. Double-check that according to the inputs you are giving, you are reading the output at the correct time and that the output is really incorrect.
- Select Show I/O Values from the View menu so you can see what all the device inputs and outputs are. After resetting the circuit, if any of the input or output values are X or Z, this usually indicates a bad connection.
- Check to see if the output is always incorrect, or if it is only wrong in certain cases. If it is only wrong in certain cases, get the circuit into a state in which the outputs are wrong, and work backward from the output to determine the source of trouble.
- Since we can assume the output is incorrect or you would not need to debug, first look at the inputs to the output gate. Are they correct? If not, figure out what they must be to be correct and write it down. Figure out how to get that value from the input gates.
- If the inputs to the output gate are correct, keep working your way backward through the circuit until you find which inputs are wrong. Most likely, you either have wires that are connected incorrectly (or not connected at all) or your design is slightly incorrect.

2.5. Scaling and printing your waveform

You can scale your timing diagram to show the level of detail you want and make it fit in the desired amount of space. To change the scale, select **Scale** from the **Options** menu you see at the top of the screen when the **Timing Diagram** is selected. Select a smaller time scale to zoom in or a larger time scale to zoom out. The 1 ns scale allows you to see the smallest time unit (1 ns) in great detail.

To fit your output on the preferred number of pages, try selecting **File**, **Print Setup**, and then select the **Landscape** option. This will print the output sideways to make it fit on fewer pages. Also, you can try selecting different scales from the **Options** menu, then checking the **Print Preview** to ensure your output will look the way you want it.

3. Creating VHDL Code

SimUaid has the capability of generating structural VHDL code that corresponds to a SimUaid circuit diagram. The VHDL code generator replaces input and output devices in your circuit diagram with input and output port signals. The architecture body of the code consists of a series of component instantiation statements. These components are defined in two VHDL packages called `SimUAid_Synthesis_Package.vhd` and `SimUAid_Simulation_Package.vhd`.

- a. First, create your design in SimUaid and save it the usual way.
- b. Test your circuit and simulate it to ensure that it works according to the specification.
- c. Add probes to all outputs. The order in which the probes are added will determine the order in which they are listed in the VHDL file.
- d. Make sure that all device labels and pin labels are valid VHDL identifiers. The only exception is that you may label pins with a single character followed by a prime (A', B', for example).
- e. Ensure that all input and output devices have labels, and that none of the labels conflict.
- f. Select the **Save VHDL** option from the file menu in SimUaid.
- g. Select **Synthesizable VHDL** from the pull-down menu if you want to generate synthesizable code. Select **VHDL for Simulation Only** if you want to preserve the delays that are associated with the SimUaid circuit and use a VHDL simulator to observe timing waveforms.
- h. Type in a valid VHDL identifier and click on **Save** to save your file.
- i. If an error message appears and the save operation fails, correct all errors according to the error messages.
- j. Repeat steps e through i as necessary until the save operation succeeds.

Your circuit will be saved as VHDL code, with all switches and input devices becoming inputs, and all probes and 7-segment indicators becoming outputs. You can use this VHDL in other programs to simulate the operation of a digital system.

4. Using N-Bit Parts, Busses, and State Machines

Using N-Bit Parts

The N-bit parts are complex parts, such as adders, registers, and counters, which have multiple data inputs and outputs. Refer to Appendix A of the *SimUaid Reference Manual* for descriptions of these parts. Typical N-bit parts have N data inputs or outputs along with other control inputs. Each data input accepts a single-bit signal and connections are made using wires. In contrast, bus parts use bus inputs and outputs for data. Each bus represents an array of N signals, and connections are made using busses instead of wires.

Example of Using N-Bit Parts

To show how to use N-Bit parts, we will construct an adder that has a register to store the sum.

- From the **Parts** menu, select **N-Bit Parts, Register**. Do not enable the **Busses** checkbox. Place the register.
- Now, select **Parts, N-bit parts, Adder**. Again, do not enable the **Busses** checkbox. Place the adder.
- Connect the adder and register and add probes and switches as shown in Figure 10. Select **Show I/O Values** on the **View** menu.

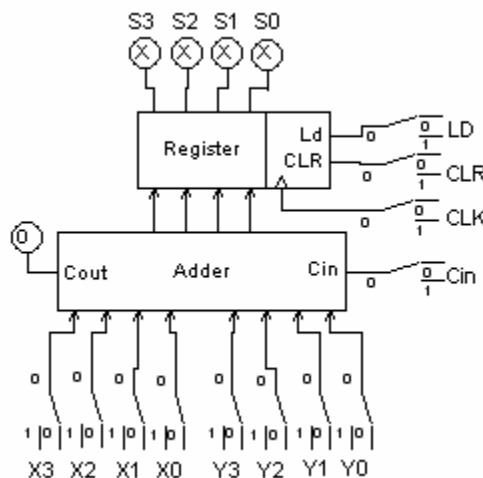


Figure 10. N-Bit Adder

- To test the adder, make sure to set the CLR switch to 0, LD to 1, and Cin to 0. Set X3-X0 to the bits of the first input and Y3-Y0 to the bits of the second addend. Notice that when

you flip the X and Y switches, the adder output changes (since it is a combinational circuit) but the register output does not change because it is clocked.

- e. Set the CLK to 0, then back to 1, and see that the adder output is loaded into the register. Notice that no matter how many times you “clock” the register (set CLK to 1 then back to 0), the sum does not change, because this adder does not have an accumulator.
- f. If you are not getting the proper result from the adder, ensure that LD, CLR, and Cin are set correctly. LD should be 1, CLR should be 0, and Cin should be 0 unless you want a carry in.

Using Busses and Bus Parts

A *bus* represents a group of wires. Connecting busses is the same as connecting wires—use the wiring tool to connect bus inputs and outputs of the same width. When you select an N-bit part from the parts menu, check "Busses" to create a bus part instead of an N-bit part. The bus parts use busses for data inputs and outputs instead of individual wires. *When you combine N-bit parts and bus parts, you have to be careful to make sure they will connect properly.* Use a bus merger to convert a group of wires to a bus, and use a bus splitter to split a bus into a group of wires. Refer to the *SimUaid Reference Manual* for how to do this.

Bus Probe and Bus Input

You use a Bus Probe to show the value on a Bus. It displays the value in hex, and the value appears on the Timing Diagram.

Use a Bus Input to assign a value to a Bus or a Bus Pin. Click on a Bus Input in LABEL mode to enter the hex string value. You cannot label Bus Input devices; you can only assign them a hex value. Enter the hex string without any leading characters, and ensure that it is of the proper length, considering the length of the bus.

Example of Using Busses

To show how to use bus parts, we will construct an adder similar to the one we created with N-bit parts.

- a. From the **Parts** menu, select **N-bit parts** and then **Register for Adder**. Enable the **Busses** checkbox. Place the register.
- b. Now, select **Parts**, **N-bit parts**, **Adder for Register**. Again, enable the **Busses** checkbox. Place the adder. Note that the Adder for Register is functionally the same as the Adder, but the adder output is at the bottom instead of the top.
- c. Connect the adder and register as shown in Fig. 11 using the wiring tool in the same way you use it to connect wires. The wiring tool can sense that you are connecting two busses, and will do this properly without you doing anything out of the ordinary.

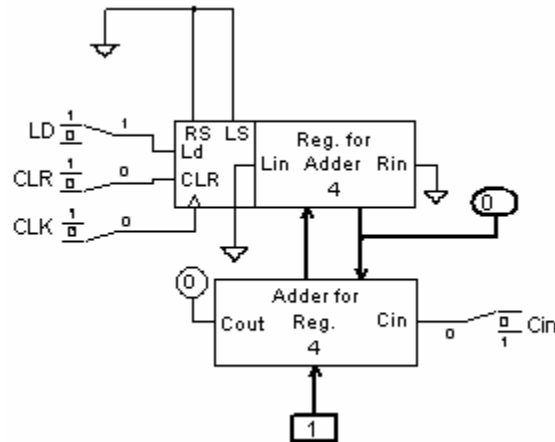


Figure 11. Bus Adder

- d. Now, select **Parts, N-bit parts, Bus Input**. Connect the bus input to the bottom input of the Adder for Register.
- e. Connect RS, LS, Lin, and Rin to ground, because we will not be right-shifting or left-shifting this time.
- f. Now, select **Parts, N-bit parts, Bus Probe**. Connect the bus probe to the output of the Register for Adder as shown in Figure 11.
- g. Add a probe and switches as shown in Figure 11.
- h. Select **Show I/O Values** on the **View** menu.
- i. To test the adder, make sure to set the CLR switch to 0, LD to 1, and Cin to 0. To set the bus input, select the labeling tool and click on the bus input. Type the hex value you want to input.
- j. To input the first value into the accumulator, set the CLK switch to 1 and then back to 0. The bus probe should show the value you just input.
- k. To add another number to the accumulator, set the bus input as desired, and then set CLK to 1 and then back to 0. The bus probe should show the sum.
- l. If you are not getting the proper result from the adder, ensure that LD, CLR, and Cin are set correctly. LD should be 1, CLR should be 0, and Cin should be 0 unless you want a carry in.

Using State Machines

You can use a Mealy state machine to control the sequence of operations in a digital system such as a multiplier or divider. The Mealy state machine, shown in Figure 12, has input pins on the lower left, and output pins on the upper left. When you change an input, the output changes asynchronously, and does not wait for the rising edge of the clock. The smaller box on the right shows the current state number. The state only changes on the rising edge of the clock. (You can simulate a Moore state machine by assigning, for each state, the same output to every arc leaving that state.)

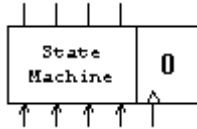


Figure 12. State Machine

Example of Using a State Machine

We will construct a simple state machine and the corresponding state graph table based on the Mealy state graph shown in Figure 13, which is the control state graph for a multiplier (see Figure 18-9 in *Fundamentals of Logic Design*, 5th Ed.)

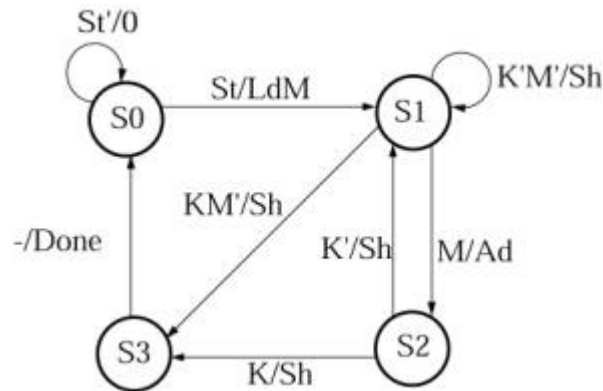


Figure 13. State Graph

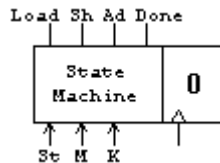
- a. To set up a state machine, first derive the state graph table from the state graph. You must have your table accessible when you create the state machine so you can use it as input to SimUaid.

- b. Each arrow in a state graph is a state transition and is represented in the table as a row. Column 1, the Present State column, contains the state name for the state that is the source of the arrow. Columns 2 and 3, the Input and Output columns, contain the signal names from the labels on the arrows. Column 4, the Next State column, contains the state name for the state that is the destination of the arrow. See Table 1 for a sample state graph table.

Table 1. State Graph Table

Present State	Input	Output	Next State
S0	St	Load	S1
S0	St'	0	S0
S1	K M'	Sh	S3
S1	K'M'	Sh	S1
S1	M	Ad	S2
S2	K'	Sh	S1
S2	K	Sh	S3
S3	-	Done	S0

- c. From the Parts menu, select Other Parts, State Machine. Change the number of inputs to 3, and then place the state machine.
- d. Next, using the labeling tool, label the state machine as shown in Figure 14.

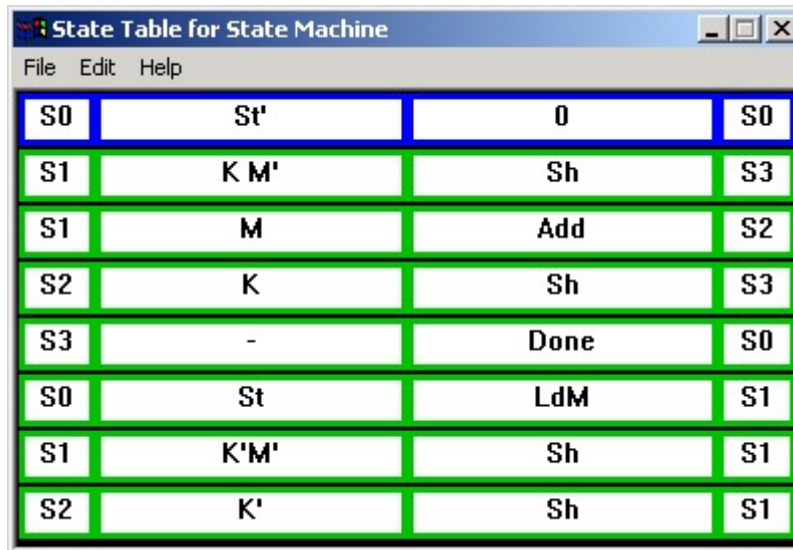
**Figure 14. Labeled State Machine**

- e. Right-click on the state machine to open the state transition table, and then input the data as shown in Table 1. SimUaid syntax-checks the data as you enter each row, so it is able to catch some typing mistakes. Each row in the table represents one arc in the state graph. For example, the second row in the table indicates that if the present state is S0 and the input is St = 1, then the output will be LdM = 1 and the next state will be S1. If two signal names appear in the same input cell, they must be separated by a space or a prime (K M' or K'M'). Similarly, spaces are required between signal names in an output cell.

The input column must be an input pin label or a combination of input pin labels ANDed together, delimited by spaces, as in, A B C. Complement the value of a pin with a prime

(A'B). Indicate a don't care value for all input pins with a dash. The output column must be an output pin label or a combination of output pin labels ANDed together, delimited by spaces, as in, Z1 Z2 Z3. An output cell containing a zero indicates that all output pins have a value of zero. Empty rows in the table are ignored. However, if a row is partially empty, you will get a syntax error.

- f. Your state graph table should look like the one in Figure 15.



Current State	Input	Output	Next State
S0	St'	0	S0
S1	K M'	Sh	S3
S1	M	Add	S2
S2	K	Sh	S3
S3	-	Done	S0
S0	St	LdM	S1
S1	K'M'	Sh	S1
S2	K'	Sh	S1

Figure 15. Resulting State Transition Table in SimUaid

- g. Reset the state machine by clicking the reset icon on the toolbar. To test the state machine, connect switches to all of the inputs. The active row in the table (the one that corresponds to the current state and input), will be shown in blue, and the other rows will be green. Note that when you change the appropriate input switch, the active row in the table changes, but the state does not. When you clock the state machine by flipping the clock input switch from 0 to 1, the active row should change to the appropriate next state.

Refer to Section 1.4.5, *State Machine*, in the *SimUaid Reference Manual* for more details about the state machine.

SimUaid Reference Manual

1. Circuit Window Menu

SimUaid follows standard MS-Windows conventions in the hierarchy of the screen and main menu. Figure 1 shows SimUaid's circuit window.

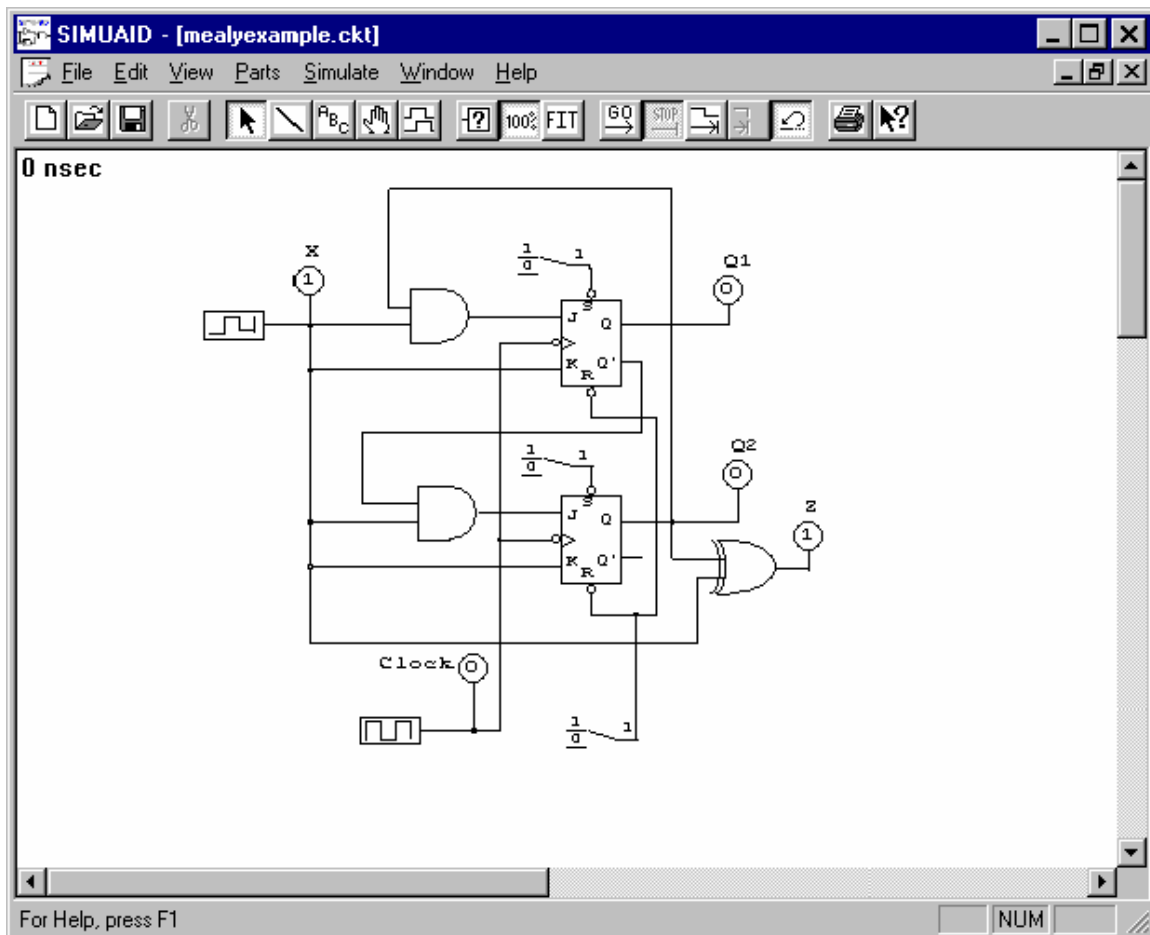


Figure 1. SimUaid Circuit Window

A menu bar is provided on the main screen with the following commands:

- **File** Allows loading, closing, saving, saving as VHDL, and printing of logic circuits.
- **Edit** Selects standard editing command—Delete, and circuit editing commands: Select, Select All, Wire, Label, Move, Delay, Default Delay, and Comment.
- **View** Turns on or off the toolbar, status bar, and all input/output values. Also selects to redraw the circuit, fit the whole circuit into window, re-size the whole circuit back to default size, set the desired scaling factor, and change the graphics mode.

- **Parts** Selects circuit parts such as gates, flip-flops, I/O devices, input signal element, checker, clock, voltage source (logic 1), and ground (logic 0), counters, registers, and other parts.
- **Simulate** Selects to run simulation step by step, go all the way to the specified simulation length, reset from start, or stop the simulation. Also provides the option to change the properties of the clock and to enter CHANGE Mode. See Appendix B for details about SimUaid modes.
- **Window** Selects window command: Cascade, Tile, Arrange Icons, and Open Timing Diagram.
- **Help** Provides access to the SimUaid help index and version information about SimUaid.

Clicking one of these commands with the left mouse button causes a pull down menu to appear. For each command, the sub-commands provided for the command are listed and may be executed by moving the cursor over the desired entry and again pressing the left mouse button. Generally, this displays a pop up window associated with this command. You can also execute some of these commands by clicking on the accelerator keys in the toolbar, or issuing short cut keys, such as Ctrl+s for saving the file. For each of the following commands, corresponding short cut keys, if any, are listed at the end of the description.

1.1. FILE MENU

The File commands are concerned primarily with loading, storing and printing of the SimUaid files. The operations provided for this command are standard and follow those found in most MS-Windows applications.

1.1.1. New

Use this command to close the currently opened SimUaid file and create a new file. SimUaid asks you to confirm saving the currently opened circuit file if it has been modified but not yet been saved. SimUaid sets the name of this circuit to Untitledx, where x is the number of new circuits you have requested.

You can open an existing logic circuit file with the **Open** command.

Shortcuts

Toolbar: 
Keys: CTRL+n

1.1.2. Open

Use this command to close the currently opened SimUaid file and open an existing file in the current window. SimUaid asks you to confirm saving the currently opened circuit file if it has been modified but not yet been saved. The **File Open dialog box** allows you to specify a file name, and the drive and directory where the file resides. A SimUaid circuit file automatically has an extension “.ckt”.

You can create a new logic circuit with the **New** command.

Shortcuts

Toolbar: 
Keys: CTRL+o

1.1.3. Save

Use this command to save the active SimUaid file to its current name and directory. When you save a document for the first time, SimUaid displays the **Save As dialog box**, so you can name your document. If you want to change the name and directory of an existing logic circuit file before you save it, choose the **Save As** command.

Shortcuts

Toolbar: 
Keys: CTRL+s

1.1.4. Save As

Use this command to save and name the active SimUaid file. SimUaid displays the **File Save As dialog box** so you can name your file.

To save a SimUaid file with its existing name and directory, use the **Save** command.

1.1.5 Save VHDL

Use this command to save the active SimUaid circuit as a VHDL description of the circuit. SimUaid displays the **Save As dialog box** so you can name the VHDL descriptor file. See the *SimUaid User's Guide* for more information.

Click on **OK** when the ‘Success in saving of VHDL file’ dialog box appears.

1.1.6. Print

Use this command to print a document. This command presents a **File Print dialog box**, where you may specify the range of pages to be printed, the number of copies, the destination printer, and other printer setup options.

The vertical and horizontal dotted, red lines you see on the circuit window are page boundaries. To keep your circuit on one page, you must fit it within the left, uppermost

rectangle. To check if your output is flowing to the next page, you can check the Print Preview from the File menu.

Shortcuts

Toolbar:



Keys:

CTRL+p

1.1.7. Print Preview

Use this command to display the active document as it would appear when printed. When you choose this command, the main window will be replaced with a print preview window in which one or two pages will be displayed in their printed format.

The **Print Preview toolbar** offers the option to view either one or two pages at a time, move back and forth through the document, zoom in and out of pages, and initiate a print job.

1.1.8. Print Setup

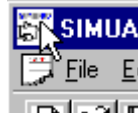
This command allows you to select the destination printer, orientation, paper size, paper source, etc.

1.1.9. Exit

Use this command to end your SimUaid session. You can also use the **Close** command on the application Control menu. SimUaid prompts you to save documents with unsaved changes.

Shortcuts

Mouse: Double-click the application's Control menu button.



Keys:

ALT+F4

1.2. EDIT MENU

The commands under the Edit menu are circuit-editing commands. These allow for cutting, selecting, wiring, labeling, moving, setting the delay for, and adding comments to circuit components.

1.2.1. Delete

Use this command to delete a currently selected circuit component. This **Delete** command is unavailable if no circuit component is currently selected. To delete a circuit component, you must be in the SELECT mode and click on a circuit component to highlight it first; then use this command to delete the selected component from the circuit.

You can delete one device, one segment, one whole *net* (interconnection of segments), or a whole circuit. Warning messages will be issued before deleting a whole net or a whole circuit.

Shortcuts

Toolbar:



Keys:

Delete key

1.2.2. Select

Use this command to switch to the SELECT mode. In the SELECT mode the mouse cursor becomes a pointer and you can select any circuit component by clicking on it. The selected circuit component will be highlighted. This function is unavailable if you are already in SELECT ALL mode.

Clicking on a wire segment highlights only that segment; however, clicking on a wire segment while holding down the Shift key highlights the entire net which contains that segment. This helps you to observe the interconnection of wires, and know which is connected to which.

When a device is selected, its label is also selected. However, when a device's label or a pin's label is selected, only the label is selected, not the entire device.

A check mark appears next to the menu item when you are in SELECT mode.

Shortcuts

Toolbar:



Keys:

CTRL+e

1.2.3. Select All

Use this command to select the whole circuit. All circuit components including devices, labels and wires are selected. A red, dotted rectangle that can just contain all the circuit components is drawn on the screen. This function is available only in SELECT and MOVE mode (i.e. you cannot use SELECT ALL when you are in WIRE or LABEL mode). To exit SELECT ALL mode, click on the background of your circuit.

A check mark appears next to the menu item when you currently select a whole circuit.

Shortcuts

Keys:

CTRL+a

1.2.4. Wire

Use this command to switch to the WIRE mode. In this WIRE mode, the mouse cursor becomes a cross, and you can connect any two points by a horizontal or a vertical segment of wire. Wires are drawn only horizontally and vertically.

You can wire different parts, nets, or busses together by clicking in the circuit window. You can draw a wire from pin to pin, from pin to wire, from wire to wire, or you can draw a dangling wire not connected to anything at all. We use the term *net* to describe a group of single, interconnected wires. A net is different from a *bus*, which can carry many signals in parallel on

the same line that you draw in SimUaid. However, you connect busses in the same way you connect wires, with the Wiring tool. For more information about connecting busses, see Section 1.4.3, *Bus Parts*.

To draw a wire, first click on a desired starting point, and then move the mouse to a desired ending point. A system beep sounds when you make a connection. The drawn wire is a combination of horizontal and vertical segments that connect the starting point and ending point. A horizontal segment is drawn from this starting point if the horizontal displacement of the ending and starting point is greater than the vertical one; otherwise, a vertical segment is drawn from this starting point. To change the direction of the initial wire segment, hold down the Shift key before clicking on the ending point. You may need to tack down the wire with intermediate clicks to force the wire go around other circuit elements, such as devices and labels.

The wire automatically terminates only when you click on a pin or wire segment. However, you can terminate a wire at any point not on a pin or wire by double clicking. If you try to connect a net to itself, a warning message appears. A dark filled circle (a dot) is drawn at intersections of more than two segments or pins to indicate a connection.

A check mark appears next to the menu item when you are in WIRE mode.

Shortcuts

Toolbar:	
Keys:	CTRL+w

1.2.5. Label

Use this command to switch to LABEL mode. In LABEL mode, the mouse cursor becomes a pencil, and you can enter a label for a device or a pin.

To label a device or a pin, first click the pencil tip on the middle of that device or pin, then enter the desired text in the on-screen edit box. You can finish editing a label by either hitting the Enter key or by clicking on anything else in the window. Clicking on any other device or pin again opens an on-screen edit box for you to enter a label for that device or pin.

A device label is placed automatically above the device. You can later move the device label wherever you wish. A pin label is placed automatically next to the pin. You can later move the pin label wherever you wish.

There is an important restriction on the labeling of input signal devices: the device label and the pin label must be the same. SimUaid will not allow you to label them with different names.

Currently, there is no restriction for the syntax of the labels. However, a warning message will be issued whenever you use the same label for two or more output pins. It asks you if you really want to connect those output pins together or not.

A check mark appears next to the menu item when you are in LABEL mode.

Shortcuts

Toolbar: 
 Keys: CTRL+l (lowercase “L”)

1.2.6. Move

Use this command to switch to MOVE mode. In MOVE mode, you can move a whole circuit, a device, a device label, or a segment.

To click and drag a circuit component, click on the component and hold down the left mouse button, then drag the marked circuit component to the desired position, and release the left mouse button to place the marked circuit component in its new position.

To move a whole circuit, first use the command **Select All** to select the entire circuit, then use the same procedure for moving a circuit component as described above to move the whole circuit. Moving the circuit out of the current view is allowed; however, it is your responsibility to scroll the view since the view will not scroll automatically when any part of the circuit is not enclosed in the view.

For a horizontal wire segment, you can only move it upward or downward; in contrast, for a vertical wire segment, you can only move it left or right. To move a horizontal wire segment, if this wire segment is connected to any vertical wire segment, the vertical wire segment is automatically lengthened or shortened vertically. Similarly, in moving a vertical wire segment left or right, if this wire segment is connected to any horizontal wire segment, the horizontal wire segment is automatically lengthened or shortened horizontally.

In moving a device, all segments attached to its input and output pins are also moved along using the same concept for moving segments described above. Connections between pins and segments still exist. You can move a device label anywhere you wish. However, that label still belongs to that device—if you move the device, both the device and its label are moved.

A check mark appears next to the menu item when you are in MOVE mode.

Shortcuts

Toolbar: 
 Keys: CTRL+m

1.2.7. Delay

Use this command to switch to the DELAY mode. In the DELAY mode, you can enter the propagation delay of any device. However, this mode is also used to select the clock rate and specify the input waveform for the input signal element.

To input the delay of a device, first click on that device, and the selected device will be highlighted in dotted lines. A **Propagation Delay dialog box** will pop up to request the delay

for that device. The allowed range of delay for any device is 1 ns to 999 ns. Click on **OK** when you are done entering the propagation delay or **Cancel** to cancel the command.

When the selected device is a clock, the propagation delay entered in the propagation delay dialog box will specify the **Clock Period** of the clock. You can change the clock period of the clock to change the speed of the clock in the circuit.

When the selected device is an input signal element, an **Input Signal dialog box** will pop up instead. This dialog box allows you to specify the waveform of the input signal element. Enter the time that the waveform changes and the new logic level; then click on **Add/Modify**. There are four available logic levels and an 'R' value. The 'R' value allows you to repeat the entire waveform from zero time. A purple line on the waveform indicates a Hi-Z and a shaded green area indicates an unknown value. An example of an input waveform is shown below. You can modify or delete any entry that you have entered. To modify an entry, double click on the entry, and the entry will appear in the time and value boxes, ready to be modified. Make the necessary modification and then click on the **Add/Modify** button. To delete an entry, highlight the entry to delete by clicking on the entry, and click on the delete button.

A check mark appears next to the menu item when you are in DELAY mode.

Shortcuts

Toolbar:



Keys:

CTRL+d

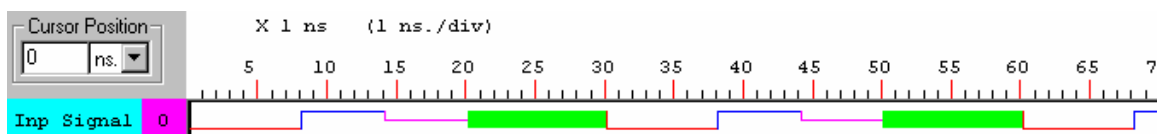
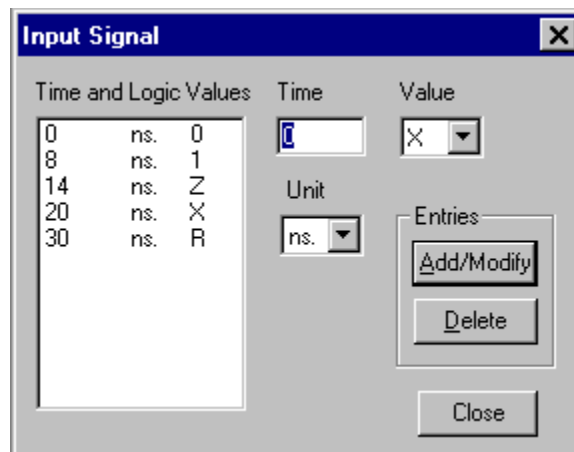


Figure 2. Specifying Input Signal

1.2.8. Default Delay

Use this command to change the default propagation delay of the device to be inserted into the circuit.

After selecting this command, a **Default Delay dialog box** allows you to specify the default propagation delay in the range 1-99 ns of the device to be inserted. After you have entered the default delay, all the devices you place on the circuit will have this propagation delay until you change the default delay again. Devices that are on the circuit before this command is executed will not be affected.

Shortcuts

Keys: ALT+e, f

1.2.9. Comment

Use this command to enter a comment that is not a label attached to a device. After selecting **Comment** from the **Edit** menu, click where you would like to start typing. To end the comment, click the mouse on another area of the screen or press the F12 key.

1.3. VIEW MENU

The commands under the View menu primarily handle the options to view the screen and circuit layout.

1.3.1. Toolbar

Use this command to display and hide the Toolbar, which includes buttons for some of the most common commands in SimUaid, such as **File Open**. A check mark appears next to the menu item when the Toolbar is displayed.



Figure 3. SimUaid Toolbar

The toolbar is displayed across the top of the application window, below the menu bar. The toolbar provides quick mouse access to many tools used in SimUaid. To hide or display the Toolbar, choose Toolbar from the View menu (ALT+v, t).

Click	Name	To
	New	Open a new SimUaid file.
	Open	Open an existing SimUaid file. SimUaid displays the Open dialog box , in which you can locate and open the desired file.
	Save	Save the active SimUaid file with its current name. If you have not named the document, SimUaid displays the Save As dialog box .
	Erase	Delete selected circuit component from the logic circuit.
	Select	Switch to SELECT mode to select circuit component.
	Wire	Switch to WIRE mode to connect circuit components.
	Label	Switch to LABEL mode to label circuit components.
	Move	Switch to MOVE mode to move circuit components.
	Delay	Switch to DELAY mode to change the propagation delay of a device.
	Show/ Hide I/O	Show or hide I/O values for all devices.
	Default Size	Resize circuit back to default size
	Fit	Fit the whole circuit into window.
	Go	Simulate current active circuit continuously.
	Stop	Stop running current simulation. However, do not reset the simulation to time zero.
	Step	Simulate current active circuit clock cycle by clock cycle.
	Change	Simulate current active circuit until a value changes
	Reset	Reset the current running simulation to time zero, reset all registers and clear the waveform.
	Print	Print the active document.
	Context Sensitive Help	Not implemented.

Table 1. Shortcut keyboard commands available in SimUaid.

Command	Shortcut
New	Ctrl + n
Open	Ctrl + o
Save	Ctrl + s
Print	Ctrl + p
Delete	Delete key
Select	Ctrl + e
Select All	Ctrl + a
Wire	Ctrl + w
Label	Ctrl + l (little L)
Move	Ctrl + m
Delay	Ctrl + d
Show I/O values	Ctrl + i
Default Size	Ctrl + v
Fit To Window	Ctrl + f
Go	Ctrl + g
Stop	Ctrl + z
Step	Ctrl + t
Change	Ctrl + c
Reset	Ctrl + r

1.3.2. Status Bar

Use this command to display or hide the Status Bar, which describes the action to be executed by the selected menu item or depressed toolbar button, and keyboard latch state. A check mark appears next to the menu item when the Status Bar is displayed. The status bar is displayed at the bottom of the SimUaid window.

The left area of the status bar describes actions of menu items as you use the arrow keys to navigate through menus. This area similarly shows messages that describe the actions of toolbar buttons as you depress them, before releasing them. If, after viewing the description of the toolbar button command you do not wish to execute the command, then release the mouse button while the pointer is off the toolbar button.

The right area of the status bar indicates which of the following keys are latched down: Caps Lock, Num Lock, and Scroll Lock.

1.3.3. Show/Hide I/O Values

Use this command to display and hide the I/O values of all devices in the current circuit. When I/O values of all devices are currently displayed, this command becomes “Hide I/O Values.” When I/O values of all gates are currently not shown, this command becomes “Show I/O Values.” Bus values are displayed in hexadecimal.

Shortcuts

Toolbar: 
 Keys: CTRL+i

1.3.4. Redraw Window

Use this command to re-draw the whole window. You might want to redraw the window occasionally if your circuit looks like it is now drawn correctly, or if your changes do not seem to take effect.

1.3.5. Default Size

Use this command to resize the whole circuit back to its default size. If the circuit already has the default size, this command is unavailable.

If the circuit is not currently at its default size, this command scales the whole circuit back to its default size. The current scrolling position is still the same.

A check mark appears next to the menu item when the circuit currently has the default size.

Shortcuts

Toolbar:



Keys:

CTRL+u

1.3.6. Fit To Window

Use this command to fit the whole circuit into the part of the window you can see.

This command scales the circuit up or down to fit it in the window, and scrolls to the position so that the whole circuit fits into the window.

Shortcuts

Toolbar:



Keys:

CTRL+f

1.3.7. Set Scaling Factor

Use this command to set the desired scaling factor for the whole circuit.

When you select this option, a dialog box appears so you can enter a scaling factor for the circuit. Select the desired scaling factor available in the pull down menu from 50% to 200%, then hit enter or click on the “OK” button.

The whole circuit is scaled appropriately using the scaling factor you enter.

1.3.8. Change Graphics Mode

Use this command to switch between the two graphics modes available in SimUaid.

When you select this option, a dialog box appears on screen to allow you to select one of the two graphics modes available in SimUaid. The modes are “Change component size” and “Same component size” mode. “Same component size” will make the components look relatively larger at higher resolutions, and relatively smaller at lower resolutions. Conversely, “Change component size” will make the components look the same size in relation to other screen elements, regardless of the resolution. Choose whichever mode works best with your operating system and computer configuration. You should close all circuit windows before changing the screen resolution.

1.4. PARTS MENU

The commands under the Parts menu allow the user to select a device from the parts library. See Appendix A for further information on N-bit parts and other parts.

These devices include:

- Basic Gates: 2 to 5 input AND, OR, NAND, and NOR
 2 input XOR and XNOR
 Inverter
- I/O Devices: Output Probe
 Switch
 7-Segment Indication
 Input Signal
- Flip-Flops: D Flip-Flop,
 J-K Flip-Flop
- N-bit Parts: Counter and Loadable Counter
 Register and Register for Adder
 Adder and Adder for Register
 N-Wide 2-to-1 MUX
 Shift Registers: Left Shift, Right Shift, and Bidirectional
 Tristate Buffers
- Bus Parts: Bus Probe and Bus Input
 Splitter and Merger
 Bus Tap
 Complementer
 Tristate Buffer
- Clock
- Checker

- +V
- Ground
- Other Parts: Full Adder, Decoder, MUX, State Machine, and Tristate Buffer

1.4.1. General

To get a device, click on the Parts menu and select the desired device. Move the mouse along and place the device in its final position by clicking the left mouse button. After you have selected a part, you can place it in the circuit as many times as you want. Double-click the left mouse button to finish placing a device and switch back to SELECT mode.

Note that for the clock, checker and state machine parts, only one of each is allowed in each circuit.

1.4.2. N-Bit Parts

The N-bit parts are complex parts that allow multiple, single inputs. If you want to use a bus input, you must use the bus parts. Refer to section 1.4.3, *Bus Parts* for detailed information.

1.4.2.1. N-Bit Adder

Use these adders to add together an array of multiple, single-bit signals. The N-bit adder and adder for register are the same, except that the adder for register has one of its sets of inputs pointing up. Refer to Appendix A, *Description of Available Parts*, for details.

1.4.2.2. N-Bit Counters

These counters count the number of transitions from 0 to 1 at their clock inputs. You can load the loadable counter in parallel when the En, LD, and CLR inputs are all 1. Refer to Appendix A, *Description of Available Parts*, for details.

1.4.2.3. N-Bit Registers

SimUaid has many different registers you can use. The basic register loads its inputs and shows them at the outputs when the right input goes from low to high. There are also three types of shift registers: Left Shift, Right Shift, and Bi-directional Shift. The Bi-directional shift register has LS and RS inputs to indicate which way to shift. If both the RS and LS inputs are 1, the Bi-directional Shift Register will function as a Left Shift Register. The register for adder is the same as the Bi-directional shift register, except that both the inputs and outputs are at the bottom. Refer to Appendix A, *Description of Available Parts*, for details.

1.4.2.4. N-Wide MUX

This MUX selects the right or left set of inputs, depending on the value of the input S. Refer to Appendix A, *Description of Available Parts*, for details.

1.4.2.5. Tristate Buffer

If the leftmost input to the Tristate Buffer is 1, the buffer acts as a set of wires, otherwise, all outputs are “Z.” Refer to Appendix A, *Description of Available Parts*, for details.

1.4.3. Bus Parts

A *bus* represents a group of wires. Connecting busses is the same as connecting wires—use the wiring tool to connect bus inputs and outputs of the same width. When you create a new bus by clicking on a bus pin or on another bus, the new bus will automatically assume the width of the pin or of the other bus.

Most of the Bus Parts have a corresponding Net Part, just as a Bus Register has a corresponding Net Register. But the following Bus Parts are related only to Bus signals. They are used to assign and show values of a Bus and to convert between a Bus and its nets.

1.4.3.1 Bus Probe and Bus Input

A Bus Probe is used to show the value of a Bus. It is analogous to the standard Probe except that it displays a hex string rather than a binary digit. It is also like the standard Probe in that each occurrence of Bus Probe in a circuit becomes a row in the Timing Diagram. A Bus Input is used to assign a value to a Bus or a Bus Pin. Click on a Bus Input in LABEL mode to enter the hex string value. The hex string is stored as the device label of a Bus Input, so it can't have a device label other than its hex string value. Enter the hex string without a leading ‘x’, and ensure that it is of the proper length.

1.4.3.2 Splitter and Merger

A Splitter is used to convert a Bus into its component Nets. It has a single input Bus Pin and a standard output pin for each one of its Nets. A Merger is used to convert a group of input signals into a Bus signal. Both of the Mergers have a single output Bus Pin. In a Net Merger all the input signals are Net signals that are merged into a Bus signal. However, for a Bus Merger, the input signals are a mix of Net signals and Bus Signals. The only constraint is that the sum of the bits in each input signal must be equal to the number of bits in the output signal.

This Bus Merger is the only device in which the width of its input pins is determined in the wiring mode. When a signal is connected to one of its input pins, the width of the signal is assigned to the width of the pin. The Bus Merger is aware of the width of its output pin and all of its input pins with a signal. When a signal is connected to its last available input pin it may give an error message indicating that sum of the input bits doesn't match the number of output bits. If there is a match, all the input signals are merged into the output signal.

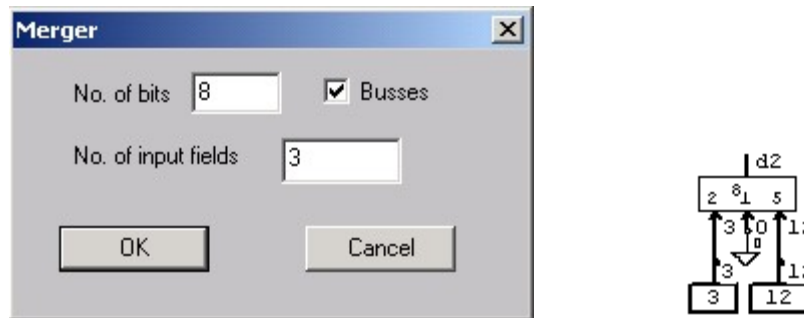


Figure 4. Bus Merger and Merger Dialog Box

Above are the dialog box and resulting Bus Merger with connecting input devices. The numbers inside a Bus Merger indicate the pin widths: the output pin is 8 wide and the input pins are 2, 1 and 5 respectively. Bus Input devices are connected to the first and third input pins with a ground on the second input pin. The result of merging the three input signals is d2 or 11010010. The first two bits are hex 3, which comes from the left Bus Input, the third bit, a 0, comes from the middle input, and the 10010 is hex 12, which comes from the right Bus Input.

1.4.3.3 Bus Tap

The Bus Tap is the only SimUaid part that you do not use the Parts Menu to place. Instead, while in the WIRE mode, hold down the Shift key, and then click on a bus to open the **Bus Tap dialog box**. Specify the **No. of bits** and select **MSB** or **LSB**. Close the dialog box and draw the tap. Double-click on the place you want to terminate the tap. You must connect a bus probe to the bus tap to see the value on the bus. Following is an example of two taps.

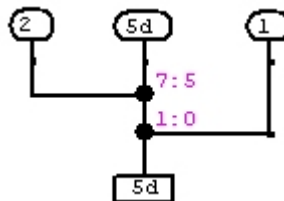


Figure 5. Bus Tap with a Width of 8

The Bus Input above has a width of 8. The three Bus Probes have widths of 3, 8, and 2.

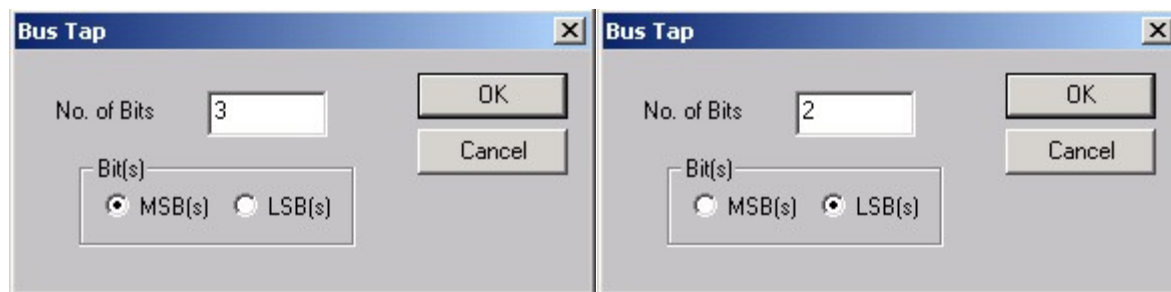


Figure 6. Bus Tap Dialog Boxes

The Bus Probe on the left is connected to the three MSBs tapped from the 8-bit Bus. The Tap Label, “7:5” indicates that bits in positions 7 to 5 are the three bits tapped.

The Bus Probe on the right is connected to the two LSB tapped from the 8 bit Bus. The Tap Label, “1:0” indicates that bits in positions 1 to 0 are the two bits tapped.

The Bus or Net drawn from a Bus Tap should terminate on an input pin or another Net or Bus.

1.4.3.4. Complementer

The Complementer outputs the hex conversion of the bitwise complement of the input string, which must also be in hex. The complement is performed once when the enable bit is set to 1. (An N-bit complementer is not available. A set of net signals can be complemented by connecting them to a merger, and then connecting the merger output to the complementer.)

1.4.4. Checker

Use this command to create a checker device. A checker device reads test data from a checker file and generates a sequence of 0’s and 1’s to test a sequential circuit. The checker device compares the output from the circuit under test with the correct output sequence and indicates PASS or FAIL.

After you select the checker, a **File Open dialog box** allows you to specify the file name, drive, and directory where the checker file resides. Checker files should have extension “.chk”.

After the checker file is successfully opened and parsed, the checker is then created. Only one checker is allowed in a circuit. Therefore, the checker option is deactivated once you have placed a checker in the circuit.

1.4.4.1. Syntax for the Checker file

The checker file has the following syntax:

<input sequence>
<output sequence>

where

<input sequence> is any combination of “1”, “0”, “X”, “R”, and space

<output sequence> is any combination of “1”, “0”, “X”, and space

X and R can be lower or upper case. X represents a don't care, and R represents a reset signal to the circuit under test. No other special characters are allowed.

Also, the number of values in both <input sequence> and <output sequence> must be equal. You can use any text editor to enter your own <input sequence> and <output sequence> to create the checker file.

1.4.5. State Machine

Use this command to place a state machine. Only one state machine is allowed in each circuit. Therefore, the state machine option is deactivated once you have placed a state machine.

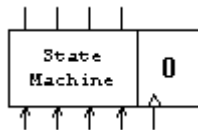


Figure 7. Basic State Machine

A state machine cycles through a predefined set of set of states according to your parameters. On its left are the input and output pins. The signals connected to these pins are asynchronous. The output changes are immediate and do not wait for the leading edge of the clock. On the right are the clock input and the display of the current state. This is the synchronous side of a state machine. The current state changes only on the rising edge of the clock.

Refer to *Example of Using a State Machine* of the *SimUaid User's Guide* for an example of how to set up a state machine.

The window for the state graph table is a floating window. When you move the window around on the screen, it floats on top of all the other windows. You cannot close the State Table window, but you can minimize it using the minus sign button on the upper right corner of the window. To show the State Table window again, click on its button on the Windows taskbar.

The state graph table has three menus: File, Edit and Help. The File menu has only the print command. Use it to print the state graph table without printing the circuit. (The state graph table is printed automatically every time the circuit is printed from the main menu.) The Edit menu has four commands: Insert Row, Delete Row, Check Syntax and Clear Table. Insert Row inserts an empty row above the currently selected row. Delete Row deletes the currently selected row. Check Syntax performs a syntax check on each cell of each row. Clear Table clears each

cell of each row. Finally, the Help menu has a single command, Colors, which documents the color changes of a row.

While the Check Syntax command checks the whole table, there are two methods of syntax checking only one cell. Each time you press the Tab key, the current cell is syntax checked before advancing to the next cell. Also, if you click on a cell, the previous cell (the cell losing the focus) is syntax checked. Finally, whenever a pin label is changed on the state machine, a full syntax check is performed on the state graph table. Remember, anytime the pin labels or the state graph table is changed, the circuit must be Reset before running it.

A special feature of a state machine is that when it is deleted, the state graph table is saved. SimUaid remembers all the cells for each row of the last state machine deleted. Although for the first state machine you place, the state graph table is empty, for all subsequent state machines that are placed, the state graph table is loaded with its previous cells. This feature is useful when you want to change the pin configuration after you have typed in the table. You may need one more output pin or one less input pin, but don't want to have to retype the table. However, if you do need an empty table at this point, use the Clear Table command in the Edit menu.

1.5. SIMULATE MENU

The commands under the Simulate menu allow you to simulate the current open circuit either step by step or continuously, to stop the simulation, and to reset the simulation. Refer to Appendix B, which explains the difference between the LIVE and CLOCK modes of SimUaid.

1.5.1. Go

Use this command to start simulating the current logic circuit and run until the time is up. Simulation continues until the time limit is up or until you use the stop command. A check mark on the simulation menu shows the current simulation mode.

Shortcuts

Toolbar: 
Keys: CTRL+g

1.5.2. Stop

Use this command to stop current running simulation. This command does not reset the logic circuit.

A check mark appears next to the menu item when you are in STOP mode.

Shortcuts

Toolbar: 
Keys: CTRL+h

1.5.3. Step

Use this command to run simulation of the current logic circuit for one clock period.

Shortcuts

Toolbar: 
 Keys: CTRL+t

1.5.4. Change

Use this command to run simulation of the current logic circuit until a value changes.

Shortcuts

Toolbar: 
 Keys: CTRL+c

1.5.5. Reset

Use this command to reset the current running simulation. It resets all devices back to their default state and sets the clock back to time zero.

Shortcuts

Toolbar: 
 Keys: CTRL+r

1.5.6. CHANGE Mode

Use this command to enter or leave the CHANGE Mode. A check appears next to the menu item when in the CHANGE Mode. This mode is only recommended for advanced users. When in the CHANGE Mode, clicking the **Change** button allows the signal to propagate through the circuit one change at a time. This is useful for debugging circuits with critical timing problems and for analyzing asynchronous circuits. When a clock is present in the circuit, the **Change** mode is not available. When no clock is present, the **Change** button is activated when a switch is toggled and signals are waiting to propagate through the circuit. See Appendix B for further discussion of simulation modes.

1.5.7. Options

Use this command to set the options for simulation, such as the length of each **Go** command and the rising- or falling-edge clock. After you select Options, an **Options dialog box** allows you to specify the simulation options.

Options dialog box

The following simulation options are available:

- **Simulation time**
 Specify the length of each **Go** simulation in ns or μ s. The default length is 100 ns, which means running continuously without stopping until the clock reaches about 100 ns. Note: when a clock is present, the simulation stops a quarter period before a clock edge.

- **Clock period**
The default clock period is 20 ns. The clock period can be changed only if there is a clock on the circuit.
- **Rising edge clock, initial value is 1**
Set initial clock output to 1. The **Step** command causes the simulator to run until a quarter period before the next rising edge of the clock.
- **Falling edge clock, initial value is 0**
Set initial clock output to 0. The **Step** command causes the simulator to run until a quarter period before the next falling edge of the clock.

1.6. WINDOW MENU

The commands under the Window menu allow you to cascade, tile the windows, and select the active window. They also allow you to arrange the icons and open the timing diagram window.

1.6.1. Cascade

Use this command to overlap all the opened windows with the active window on the top.

1.6.2. Tile

Use this command to see all the opened windows on one screen.

1.6.3. Arrange icons

Use this command to arrange the icons of the minimized windows.

1.6.4. Open/Close Timing Diagram

SimUaid will open another window to display the timing diagram formed from a simulation of the circuit. To close the timing diagram window, click on the window with the circuit, and then choose **Close Timing Diagram** from the Window menu.

The timing diagram will be empty unless you add probes to your circuit and then create waveforms by simulating your circuit. The order of the waveforms in the timing diagram is the same as the order in which you connect the probes. To clear the timing diagram, select **Reset** from the Simulate menu.

1.7. HELP MENU

The commands under the Help menu allow you to get help and references for the SimUaid software.

1.7.1. SimUaid Help Index

Use this command to display the opening screen of Help. From the opening screen, you can jump to step-by-step instructions for using SimUaid and various types of reference information.

1.7.2. Using Help

Use this command for instructions about using Help.

1.7.3. About

Use this command to display the copyright notice and version number of your copy of SimUaid.

1.7.4. Context Sensitive Help

This is on the toolbar, and not in the Help Menu. Not implemented in this version.

Shortcut

Toolbar:



Keys:

SHIFT+F1

2. TIMING WINDOW MENU

Figure 9 shows the SimUaid timing window for the circuit in Figure 1. This window displays the waveform of the circuit drawn in the circuit window.

Place a probe at any connection you wish to observe and the waveform at the probe will be plotted in the timing window. The order of the waveforms in the timing window is determined by the order in which the probes are placed. The waveform in the timing window takes the label name of the probe it is plotting. To delete a waveform from the timing window, you will have to delete the corresponding probe in the circuit diagram.

You can place the cursor anywhere on the waveform by clicking on the waveform itself. The cursor box will always display the current position of the cursor. Another way to position the cursor is to type in the cursor box the exact position where you want the cursor and press enter. The value of the waveform at the cursor is displayed beside the waveform. And every time the cursor position changes, the value is updated.

The tool bar of the timing window is the same as the tool bar for the circuit window but with some of the functions disabled.

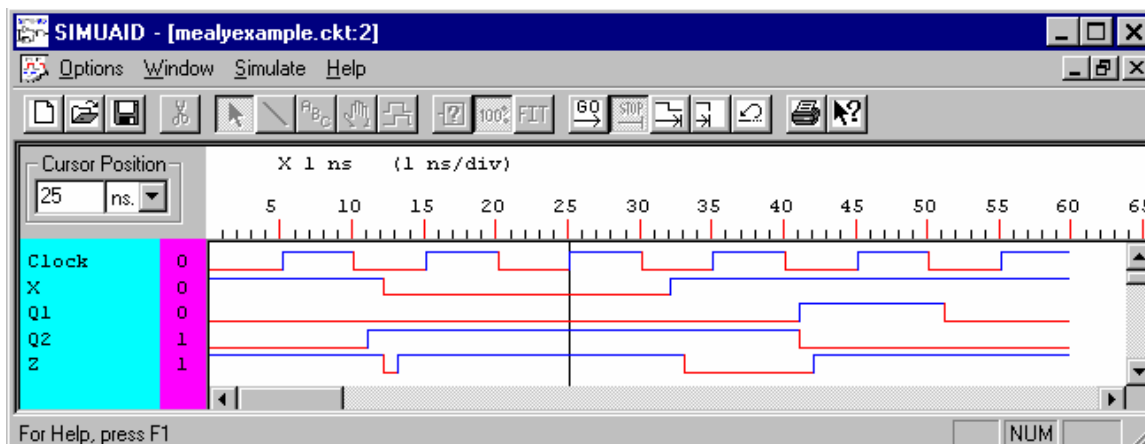


Figure 9. SimUaid Timing Diagram Window for the Circuit in Figure 1.

A menu bar is provided on the main screen with the following commands:

- **Options** Allows scaling, closing, and printing of timing diagram.
- **Window** Selects window command: Cascade, Tile, and Arrange icons. Refer to the Window menu in the circuit window for more detail.
- **Simulate** Selects running simulation step by step, gives the specified simulation length, resets from start, or stops the simulation. This menu item is the same as the Simulate menu in the circuit window. Refer to the Simulate menu in the circuit window for more detail.
- **Help** Provides the context sensitive help. Describes operation of the current screen as well as other portions of the software for on line reference. Refer to the Help menu in the circuit window for more detail.

2.1. Options Menu

The commands under the Options menu allow the user to choose the scale, print the waveform and close the timing diagram window.

2.1.1. Scale

Use this command to change the scale of the waveforms. Select a smaller time scale to zoom in or a larger time scale to zoom out. The 1 ns scale allows you to see the smallest time unit (1 ns) in great detail.

To fit your output on the preferred number of pages, try selecting File, Print Setup, then select the Landscape option. This will print the output sideways to make it fit on fewer pages. Also, you can try selecting different scales from the Options menu, then checking the Print Preview to ensure your output will look the way you want it.

2.1.2. Print, Print Preview and Print Setup

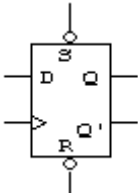
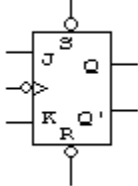
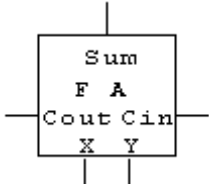
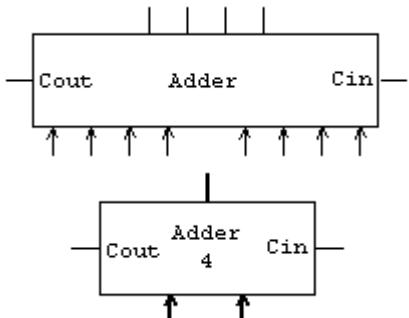
Refer to the print, print preview, and print setup commands in the circuit menu for details.

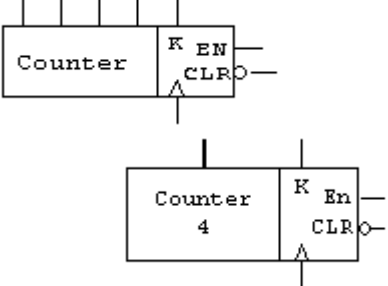
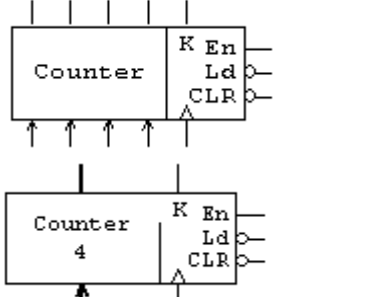
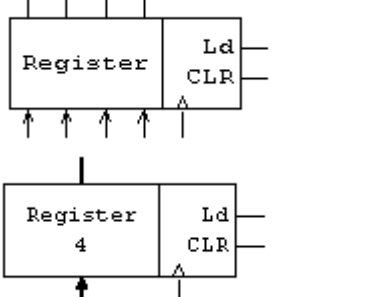
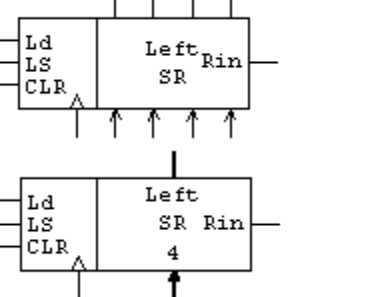
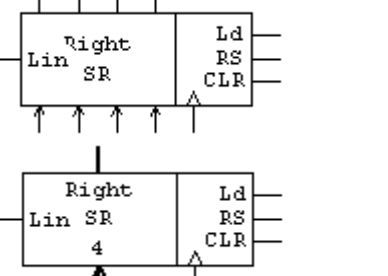
2.1.3. Close

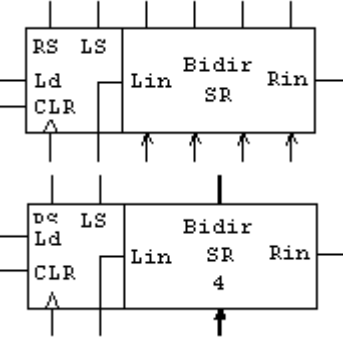
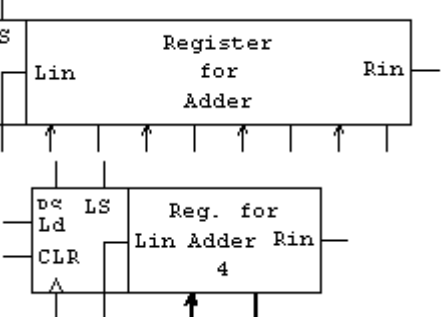
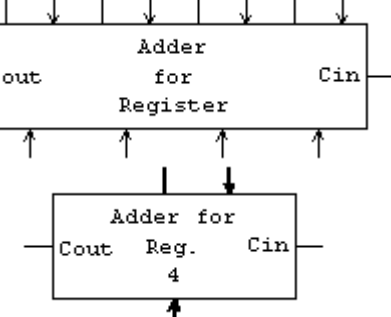
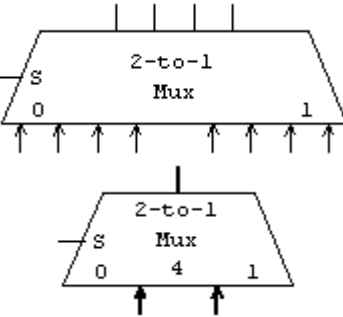
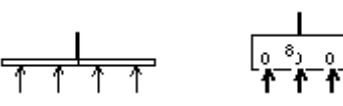
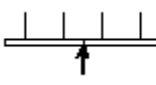
Use this command to close the timing window. Closing the timing window will not destroy the waveform. To clear the waveform you will have to reset the simulation.

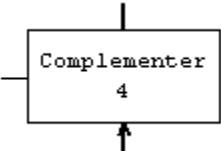
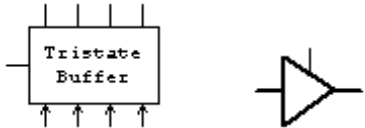
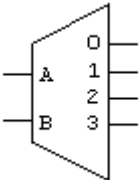
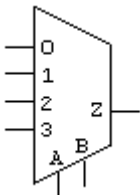
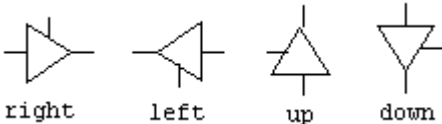
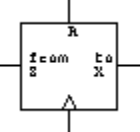
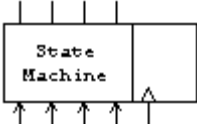
Appendix A: Description of Available Parts

Note: All parts with clock input change state on rising edge of the clock (except J-K flip-flop). Except for flip-flops, all parts with a clock input are fully synchronous, which means the output only changes in response to the clock.

	7-Segment: Segment i is on when the corresponding input is 1. 
	Input Signal: See section 1.2.7.
	Clock: To set the clock period, type Ctrl+d, then click on the clock symbol on the circuit. Only one clock per circuit is allowed.
	D Flip-Flop: The set and reset inputs are active low and override the clock input.
	J-K Flip-Flop: The set and reset inputs are active low and override the clock input. Note that state change occurs on falling edge of clock.
	Full Adder: Adds X, Y, and Cin to give Sum and Cout.
	N-bit Adder: (2 to 16 bits) Adds left N-bit input, right N-bit input, and Cin to give an N-bit Sum and Cout. Bus N-bit Adder: (2 to 32 bits)

	Counter: (2 to 16 bits, fully synchronous) Counter is cleared when $\overline{CLR} = 0$, Increments when $\overline{CLR} = 1$ and $En = 1$. $K = 1$ when all counter output bits are 1. Bus Counter: (2 to 32 bits, fully synchronous)
	Loadable Counter: (2 to 16 bits) Same as counter, but loads when $Ld = 0$ and $\overline{CLR} = 1$. Increments when $En = 1$, $\overline{Ld} = 1$, and $\overline{CLR} = 1$. Bus Loadable Counter: (2 to 32 bits)
	Register: (2 to 16 bits) Clears when $CLR = 1$, Loads when $CLR = 0$ and $Ld = 1$. Bus Register: (2 to 32 bits)
	Left Shift Register: (2 to 16 bits) Clears when $CLR = 1$, Loads when $CLR = 0$ and $Ld = 1$, Shifts left when $CLR = Ld = 0$, $LS = 1$ (data on right input pin is shifted in). Bus Left Shift Register: (2 to 32 bits)
	Right Shift Register: (2 to 16 bits) Clears when $CLR = 1$, Loads when $CLR = 0$ and $Ld = 1$, Shifts right when $CLR = Ld = 0$, $RS = 1$ (data on left input pin is shifted in). Bus Right Shift Register: (2 to 32 bits)

	Bi-directional Shift Register: (2 to 16 bits) Similar to Left Shift Register and Right Shift Register, except shifting either way is possible. Bus Bi-directional Shift Register: (2 to 32 bits)
	Register for Adder: (2 to 16 bits) Same as Bi-directional Shift Register except both input and output pins are at bottom. Bus Register for Adder: (2 to 32 bits)
	Adder for Register: (2 to 16 bits) Same as N-bit adder, except one set of inputs is at the top. Bus Adder for Register: (2 to 32 bits)
	N-Wide 2-to-1 Mux: (2 to 16 bits) Selects the left set of inputs when S = 0; selects the right set of inputs when S = 1. Bus N-Wide 2-to-1 Mux: (2 to 32 bits)
	Merger: (2 to 32 bits) Merges signals into a single Bus signal. May be connected to an input bus pin or a bus. In a Bus Merger, the number below the output pin indicates its number of bits. The numbers above the input pins indicate their number of bits.
	Splitter: (2 to 32 bits) Splits a bus signal into single bit signals. It converts a Bus to its component Nets. May be connected to an output bus pin or a bus.

	Bus Input: (2 to 32 bits) Inputs the value of a bus as a hex string. May be connected to an input bus pin or a bus. To input a bus value: enter label mode and click on the Bus Input.
	Bus Probe: (2 to 32 bits) Displays the value of a bus as a hex string. May be connected to an output bus pin or a bus.
	Complementer: (2 to 32 bits) Performs the 1's complement of its input. The pin on the left enables or disables the Complementer.
	N-bit Tristate Buffer: (Net 2 to 16 bits, Bus 2 to 32 bits) When the control input is 0, the buffer outputs are high-Z; when it is 1, the outputs are the same as the inputs.
	Decoder: (2 to 4, or 3 to 8) If AB = 00, output 0 is 1; if AB = 01, output 1 is 1; if AB = 10, output 2 is 1; etc.
	Mux: (2-to-1, 4-to-1, or 8-to-1) AB inputs select the input pin that is routed to the output. If AB = 00, Z = input 0; if AB = 01, Z = input 1; etc.
	Single-bit Tristate Buffer: A Tristate Buffer is used to turn a signal off and on.
	Checker: A checker device reads test data from a checker file and generates a sequence of 0's and 1's to test a sequential circuit. Refer to Section 1.4.4, <i>Checker</i>, for more information about the checker.
	State Machine: (1 to 8 inputs, 1 to 8 outputs) A state machine cycles through a predefined set of set of states according to your parameters. Right-click on the state machine to enter the associated state table.

Appendix B: SimUaid Modes of Operation

Live Mode

When a new circuit window is opened, the simulator is in LIVE mode. In this mode, signals propagate through the circuit in a manner similar to an actual circuit. When placing components, inputs are assigned a value of "Z" (high-impedance), and device outputs may be "X" (unknown) until they are reset or changed. As soon as a signal is connected to a gate input, that signal propagates through the gate, and if the gate output changes, the change is propagated to the next device input. Such changes continue to propagate through the circuit until no further output changes occur (or until the simulation time limit is reached).

In LIVE mode, if the inputs to the circuit come only from switches, using the timing diagram window is not recommended because time does not advance in the normal manner. Use probes and “show I/O values” instead of displaying waveforms. Whenever a switch is toggled, the resulting signal change propagates through the circuit until either no further change occurs or the simulation time limit is reached. The **Reset** button on the toolbar is active in live mode, but the **Go**, **Step**, and **Change** buttons are normally disabled unless a clock or input device is present. If a clock is present, the simulator remains in live mode until the **Step** or **Go** button is clicked.

Clock Mode (Synchronous Mode)

This mode is intended for testing synchronous sequential circuit in which state changes occur on the rising or falling edge of the clock. When a clock is placed on the circuit diagram, the **Go** and **Step** buttons become active. Clicking one of these buttons puts the simulator in clock mode. Each time **Step** is clicked, the simulator runs until a quarter period before the next rising edge of the clock (or falling edge if this has been selected in the **Simulation Options dialog box**). For example, if the clock period is 20 ns, the simulator will run for 15 ns. This feature is very useful for debugging synchronous circuits since the user can check all device inputs a short time before the active edge of the clock. Each time **Go** is clicked, the simulator runs for about the simulation time limit, but the simulator stops a quarter period before a rising (or falling) edge of the clock. Thus, if the circuit is reset, the clock period is 20 ns, and the simulation time limit is 100 ns, clicking **Go** causes the simulator to run for 95 ns.

Change Mode

CHANGE mode is intended for advanced users. It allows you to step through the circuit one output change at a time. CHANGE mode is mainly intended for simulating asynchronous circuits. After a circuit has been entered and reset, you can enter change mode by clicking on **Change Mode** on the Simulate menu. After entering the change mode, you should follow the instructions that appear at the bottom of the window. The **Change** and **Go** buttons on the toolbar will then become active whenever any changes are pending at the device outputs. The simulator uses an event queue to keep track of pending changes. For example, if the input to an

inverter changes to 1 at time 4 ns, and the inverter has a delay of 2 ns, then the output of the inverter will change to 0 at time 6 ns, and this pending change is put in the event queue. You must toggle a switch to put pending change(s) into the event queue. Each time you click the **Change** button, a pending change is removed from the queue and that change propagates to the next device input(s). This, in turn could cause more pending change(s) to be put into the event queue. Clicking **Go** propagates all changes through the circuit until no further changes are pending or until the simulation time limit is reached. Clicking **Reset** removes all pending changes from the queue in addition to resetting time to 0 and clearing the waveforms. However, this may leave the circuit in a strange condition with unresolved device output changes. Therefore, we recommend that you use **Change** or **Go** to remove pending changes before resetting the circuit.

If a clock is present, **Change Mode** is not available. In the present version of SimUaid, Change Mode should not be used when an input device is present

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